

AHMED FERGANEY

Digital Transformation Engineer

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PROFILE

Manufacturing engineer bridging shop-floor process improvement with digital and data-driven tools, across automotive and home appliance production environments. Combines hands-on Lean/Six Sigma process expertise with Python, SQL, statistical analysis, embedded systems, and automation/IoT foundations to support smart-manufacturing initiatives, data-backed decision-making, and cross-functional digital transformation projects.

CORE QUALIFICATIONS

Data & Digital Manufacturing: Python for data analysis, SQL for manufacturing/production data querying, Statistical Process Control (SPC, Cp/Cpk, control charts, hypothesis testing), Power BI dashboards & data visualization

Automation, IoT & Smart Manufacturing: PLC fundamentals & Classic Control | Embedded Linux (Yocto, Buildroot), ARM, SPI/I²C/UART/CAN | Python, C/C++, FastAPI, REST APIs | SQL, Database Design, Event-Driven Architecture, RabbitMQ | Model Deployment & Inference APIs

Process Engineering & Continuous Improvement: Process design, mapping & optimization (VSM, SIPOC), line balancing & capacity planning, work standardization & SOP development, Lean Manufacturing (Kaizen, 5S, SMED), Six Sigma fundamentals (DMAIC), KPI management (OEE, FPY, scrap rate, downtime)

Quality & Standards: 8D, RCA, CAPA, PFMEA/DFMEA, Pareto analysis, audit participation & compliance processes

Leadership: Cross-functional Team Leadership, Training & Process Transformation Initiatives (aligned with DevOps/MLOps practices), Technical Mentoring, Agile/Scrum Methodologies

PROFESSIONAL EXPERIENCE

Product Development Engineer | Kandil Glass | 10th of Ramadan City, Egypt06/2025 – Present

- Led end-to-end development of customized glass products, from initial client requirements (via sales) through feasibility analysis, design, and full-scale manufacturing.
- Monitored production performance and analyzed data to optimize quality, efficiency, and delivery across design, production, and operations.

Operational Excellence Engineer | Midea Electric | 6th of October City, Egypt02/2025 – 05/2026

- Led cross-functional training and process transformation initiatives, enabling knowledge scaling and collaboration aligned with DevOps and MLOps practices.
- Led a work standardization (Standard Work) project across production lines, improving Line of Balance (LOB) from 60% to 75%, using structured data analysis to identify workload imbalances — within the Home Appliances sector.
- Directly supervised and coached a production/process team, embedding data-informed decision-making into daily performance reviews.

Process Quality Engineer | Methode Electronics | Nasr City, Egypt12/2023 – 12/2024

- Reduced defect rate from 8% to 2% (75% reduction) on the VW automotive assembly line by applying root cause analysis (RCA), PFMEA, and corrective actions, supported by structured data tracking.
- Applied root cause analysis (RCA) and structured problem-solving methodologies (DMAIC, DFMEA) to diagnose and resolve failures, strengthening debugging capabilities transferable to distributed systems and API reliability.
- Recorded and managed inspection results and quality documentation using the site's ERP/QMS system, supporting traceability and digital record-keeping.

Continuous Improvement Engineer | JAC | Badr City, Egypt03/2023 – 11/2023

- Increased Water Dispenser line productivity from 150 to 225 units per 2-hour cycle (50% improvement) through data-driven process and workflow optimization.
- Held continuous improvement responsibility for the Water Dispenser and TV production lines.

EDUCATION

B.Sc. in Mechanical Engineering | Banha University | 2016 – 2022

- Academic Standing: Ranked 5th out of 120 in the Mechanical Engineering Department
- Graduation Project: Design, Modeling, and Control of a Six-DOF Artificial Upper Limb

CERTIFICATIONS

- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, AI Agents | Udemy | 2026
- Prompt Engineering and RAG | Eng. Abu Bakr Soliman | 2026
- Fundamentals of Backend Engineering | Udemy | 2025
- FastAPI Development | freeCodeCamp | 2025
- Database, Cloud Computing, Kubernetes, Jenkins, and Redis | Udemy | 2025
- CNN, RNN, LSTM, GRU, and Transformers | Stanford University | 2024
- Embedded Linux, CMake, Google Test, GDB, QEMU | Eng. Moatasem Elsayed | 2024
- Embedded System Diploma | Eng. Ahmed Abd El-Ghafar | 2023
- Lean Six Sigma Yellow and Green Belt | Udemy | 2022
- Robotics Camp with a Case Study (Robotic Arm) | RTE | 2020
- Mechanics of Automotive Diploma | Don Bosco Institute | 2021
- Automation Basic Diploma | HA Consulting Group | 2020
- Turning Course | Don Bosco Institute | 2020
- Oxyacetylene and ARC-Welding in High-Pressure Pipes Diploma | Don Bosco Institute | 2019

LANGUAGES & TOOLS

Languages: Arabic (Native), English (Professional), German (Basic)

Tools: Python, SQL, FastAPI, Docker (docker.com/u/ahmedferganeey), Power BI, MS Office, Jira, Trello, MATLAB, Packet Tracer, Cursor, STM32CubeMX